UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 00971

LH RADIATION AND RELATIVITY

SUMMER EXAMINATION 2024

Time Allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and a table of physical constants and units that may be required will be found at the end of this question paper.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. In this question we consider fields of 4-force f^μ acting on a relativistic particle of rest mass m, 4-velocity u^μ and 4-acceleration α^μ. Assume the relativistic force law $f^\mu = m\alpha^\mu$ always holds.

   (a) Prove Minkowski's equation of motion,

   $$f^\mu u_\mu \;=\; 0.$$

   State whether f^μ is spacelike, timelike or null, and interpret this property physically. **[6]**

   (b) Prove that a 4-force of the form $f^\mu = a^{\mu\nu}u_\mu$ always satisfies Minkowski's equation of motion, assuming $a^{\mu\nu}$ is an antisymmetric tensor. Show that 3-vector form of the Lorentz force law corresponds to such a force, with $a^{\mu\nu}$ equal to the Faraday tensor $F^{\mu\nu}$ times the particle rest charge q. **[6]**

   (c) We now explore how to construct a 4-force from a Lorentz scalar potential field $V(x^\mu)$,

   $$f^\mu \;=\; \partial^\mu V - \frac{1}{c^2}\frac{\mathrm{d}}{\mathrm{d}\tau}(Vu^\mu).$$

   Explain how this f^μ is tensorial. Prove that this 4-force satisfies Minkowski's equation of motion. **[9]**

   (Hint: consider the relativistic chain rule when taking the τ derivative of $V(x^\mu)$.

   (d) Assuming that $\partial_t V = 0$, show that $f^\mu = m\alpha^\mu$ for the 4-force in (c) can be rewritten

   $$\begin{aligned}\gamma\frac{\mathrm{d}}{\mathrm{d}t}(m\gamma c) &= -\frac{\gamma}{c}\frac{\mathrm{d}}{\mathrm{d}t}(\gamma V)\\ \gamma\frac{\mathrm{d}}{\mathrm{d}t}(m\gamma \boldsymbol{v}) &= -\boldsymbol{\nabla}V - \frac{\gamma}{c^2}\frac{\mathrm{d}}{\mathrm{d}t}(\gamma V\boldsymbol{v}).\end{aligned}$$

   Show that, in the nonrelativistic regime $v \ll c, V \ll mc^2$, these expressions reduce to nonrelativistic equations involving a Newtonian scalar potential energy $V(\boldsymbol{r})$. **[9]**

2. (a) Using the definition of the Faraday tensor $F^{\mu\nu}$ in terms of the 4-potential, and using relationships between the field components and potentials (see Formula sheet), show that the dual Faraday tensor $G^{\mu\nu}$ satisfies **[6]**

$$G^{0i} \;=\; B^i \qquad \text{and} \qquad G^{ij} \;=\; -\varepsilon^{ijk} E_k/c.$$

(b) Write down the Lorentz transformation of the dual Faraday tensor $G_{\overline{\mu\nu}}$ in frame $\overline{\Sigma}$ in terms of components in Σ. Assuming that these frames are in standard configuration, derive the Lorentz transformation rules for the magnetic field components $\overline{B}_x$ and $\overline{B}_y$. **[8]**

A very long thin cylindrical wire with electric charge density ρ and small radius a lies at rest along the x-axis of the lab frame Σ. According to Gauss's law, this charge creates a transverse electric field, given at a perpendicular distance, $r_\perp > a$ from the wire by

$$\boldsymbol{E}_\perp \;=\; \frac{a^2\rho}{2\varepsilon_0 r_\perp}\frac{\boldsymbol{r}_\perp}{r_\perp}.$$

An observer moving parallel to the wire at speed v detects both an electric field, $\overline{\boldsymbol{E}}$, and a magnetic field, $\overline{\boldsymbol{B}}$

(c) Use the field transformation laws from the formula sheet and the expression for $\boldsymbol{E}_\perp$ in Σ, given above, to find $\overline{\boldsymbol{E}}$ and $\overline{\boldsymbol{B}}$ in the observer's frame $\overline{\Sigma}$. **[4]**

(d) Show that the same results can be obtained in a different way: write down the current density 4-vector j^μ in Σ, then Lorentz transform it to $j^{\overline{\mu}}$ in $\overline{\Sigma}$. Thus find $\overline{\boldsymbol{E}}$ and $\overline{\boldsymbol{B}}$ in $\overline{\Sigma}$ using the integral form of the inhomogeneous Maxwell equations (cf. the formula sheet). **[12]**

3. (a) The radiated electromagnetic field of an accelerated charge q is given by

$$\boldsymbol{E}_{\rm rad} \;=\; \frac{1}{4\pi\varepsilon_0}\left[\left[\frac{q\boldsymbol{e}_R\times\{(\boldsymbol{e}_R-\boldsymbol{\beta})\times\boldsymbol{a}\}}{Rc^2(1-\boldsymbol{e}_R\cdot\boldsymbol{\beta})^3}\right]\right]$$

and $\boldsymbol{B} = c^{-1}\left[[\boldsymbol{e}_R]\right]\times\boldsymbol{E}$, where $R^\mu = x^\mu - x^\mu_{\rm ret} = (R, R\boldsymbol{e}_R)$ is the null vector between the observation event and the retarded event at which the energy was radiated by the charge. Explain by means of a diagram or otherwise what the various terms in the expression mean, and identify the properties making this a radiation field. **[8]**

(b) Calculate the energy density $W_{\rm rad}$ and the Poynting vector $\boldsymbol{S}_{\rm rad}$ of the relativistic radiation field. How are their magnitudes related? Show that

$$|\boldsymbol{E}_{\rm rad}|^2 \;=\; \frac{q^2}{(4\pi\varepsilon_0)^2R^2c^4}\frac{|(\boldsymbol{e}_R-\boldsymbol{\beta})\times\boldsymbol{a})|^2-[\boldsymbol{e}_R\cdot(\boldsymbol{a}\times\boldsymbol{\beta})]^2}{(1-\boldsymbol{e}_R\cdot\boldsymbol{\beta})^6}.$$

(Hint: Recall that for two 3-vectors, $|\boldsymbol{b}\times\boldsymbol{d}|^2 = |\boldsymbol{b}|^2|\boldsymbol{d}|^2-(\boldsymbol{b}\cdot\boldsymbol{d})^2$.) **[5]**

(c) Consider the retarded spacetime point $x^\mu_{\rm ret}(\tau) = [[x^\mu]] = (ct_{\rm ret}, \boldsymbol{r}_{\rm ret})$ at which the energy was radiated by the charge. Show that

$$\partial_\mu x^\nu_{\rm ret} \;=\; u^\nu\partial_\mu\tau \;=\; \frac{u^\nu R_\mu}{R^\sigma u_\sigma},$$

where τ is the proper time experienced by the charge, and u^μ denotes the 4-velocity at retarded time. Hence show that **[7]**

$$\frac{\partial t_{\rm ret}}{\partial t} \;=\; \frac{1}{1-\boldsymbol{e}_R\cdot\boldsymbol{\beta}}.$$

(d) Write down an expression for the Poynting vector $\boldsymbol{S}_{\rm rad}$ assuming both $\boldsymbol{\beta}$ and $\boldsymbol{a}$ are parallel to the z-direction. Consider the total energy radiated at retarded time within a thin spherical shell $\mathcal{S}$ of radius R centred at $\boldsymbol{r}_{\rm ret}$,

$$\frac{{\rm d}\mathcal{E}_{\rm rad}}{{\rm d}t} \;=\; \int_{\mathcal{S}}(\boldsymbol{S}_{\rm rad}\cdot\boldsymbol{e}_R)R^2\sin\theta{\rm d}\theta{\rm d}\phi.$$

Explain how you can use your result from (c) to get an expression for the relativistic Larmor formula for the energy radiated at retarded time, for parallel velocity and acceleration, **[10]**

$$\frac{{\rm d}\mathcal{E}_{\rm rad}}{{\rm d}t_{\rm ret}} \;=\; \frac{q^2a^2\gamma^6}{6\pi\varepsilon_0c^3}$$

You may quote the following integral without proof:

$$\int_0^\pi\frac{\sin^3\theta}{(1-b\cos\theta)^5}{\rm d}\theta \;=\; \frac{4}{3}\frac{1}{(1-b^2)^3}.$$

Formula Sheet

- **Maxwell's equations** in 3-form, for electric field $\boldsymbol{E}$, magnetic field $\boldsymbol{B}$, electric current density $\boldsymbol{J}$ and charge density ρ, in empty space, with electric constant ε_0 and magnetic constant μ_0, where $c = 1/\sqrt{\varepsilon_0\mu_0}$,

$$\begin{aligned} \nabla\cdot\boldsymbol{B} &= 0, & \nabla\times\boldsymbol{E}+\dot{\boldsymbol{B}} &= 0, \\ \varepsilon_0\nabla\cdot\boldsymbol{E} &= \rho, & \mu_0^{-1}\nabla\times\boldsymbol{B}-\varepsilon_0\dot{\boldsymbol{E}} &= \boldsymbol{J}, \end{aligned}$$

- **Lorentz force equations** in in 3-form, for particle charge q, speed $\boldsymbol{v}$, momentum $\boldsymbol{p}$, and energy $\mathcal{E}$,

$$\frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t} = q(\boldsymbol{E}+\boldsymbol{v}\times\boldsymbol{B}), \qquad \frac{\mathrm{d}\mathcal{E}}{\mathrm{d}t} = q\boldsymbol{E}\cdot\boldsymbol{v}.$$

- **Lorentz transformed EM fields** after a Lorentz boost by velocity $\boldsymbol{v}$,

$$\begin{aligned} \overline{\boldsymbol{E}} &= \gamma(\boldsymbol{E}+\boldsymbol{v}\times\boldsymbol{B}) - \frac{\gamma^2}{c^2(\gamma+1)}\boldsymbol{v}(\boldsymbol{v}\cdot\boldsymbol{E}), \\ \overline{\boldsymbol{B}} &= \gamma(\boldsymbol{B}-c^{-2}\boldsymbol{v}\times\boldsymbol{E}) - \frac{\gamma^2}{c^2(\gamma+1)}\boldsymbol{v}(\boldsymbol{v}\cdot\boldsymbol{B}) \end{aligned}$$

- **Electromagnetic potentials** with V the scalar potential, $\boldsymbol{A}$ the vector potential,

$$\boldsymbol{B} = \nabla\times\boldsymbol{A}, \qquad \boldsymbol{E} = -\dot{\boldsymbol{A}} - \nabla V.$$

- **Minkowski Metric**, given by $\eta_{\mu\nu}$ or $\eta^{\mu\nu}$, where all components zero except

$$\eta_{00} = \eta^{00} = +1, \qquad \eta_{jj} = \eta^{jj} = -1, \qquad \text{for } j = 1,2,3.$$

- **Spacetime position** x^μ transforms as contravariant 4-vector,

$$x^\mu = (x^0, x^1, x^2, x^3) = (ct, \boldsymbol{r}); \qquad x_\mu = (x_0, x_1, x_2, x_3) = (ct, -\boldsymbol{r}).$$

- **Covariant Derivative**, ∂_μ transforms as covariant 4-vector,

$$\partial_\mu = (\partial_0, \partial_1, \partial_2, \partial_3) = (c^{-1}\partial_t, \boldsymbol{\nabla}), \qquad \partial^\mu = (\partial^0, \partial^1, \partial^2, \partial^3) = (c^{-1}\partial_t, -\boldsymbol{\nabla}).$$

- **Lorentz transformation**, $\Lambda^{\overline{\mu}}_{\nu}$ from Σ to $\overline{\Sigma}$, for a boost by velocity $v = \beta c$ in standard configuration, has nonzero components

$$\Lambda^{\overline{0}}_{0} = \Lambda^{\overline{1}}_{1} = \gamma, \quad \Lambda^{\overline{0}}_{1} = \Lambda^{\overline{1}}_{0} = -\gamma\beta, \quad \Lambda^{\overline{2}}_{2} = \Lambda^{\overline{3}}_{3} = 1,$$

with the Lorentz factor $\gamma = \gamma(v) = (1-\beta^2)^{-1/2}$.

- **d'Alembertian operator**, a Lorentz scalar operator,

$$\Box^2 \;\equiv\; \eta^{\mu\nu}\partial_\mu\partial_\nu \;=\; c^{-2}\partial_t^2 - \nabla^2.$$

- **Faraday tensor** in matrix form

$$F_{\mu\nu} \;=\; \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{pmatrix},$$

$$F^{\mu\nu} \;=\; \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix},$$

so $F_{0i} = E_i/c$ and $F^{ij} = -\varepsilon^{ijk}B_k$, for $i, j, k = 1, 2, 3$.
It follows from the 4-potential $A^\mu \;=\; (V/c, \boldsymbol{A})$ that

$$F_{\mu\nu} \;=\; \partial_\mu A_\nu - \partial_\nu A_\mu.$$

- **dual Faraday tensor** $G_{\mu\nu}$ is given by

$$G_{\mu\nu} \;=\; \tfrac{1}{2}\varepsilon_{\mu\nu\alpha\beta}F^{\alpha\beta}, \qquad G^{\mu\nu} \;=\; \tfrac{1}{2}\varepsilon^{\mu\nu\alpha\beta}F_{\alpha\beta}.$$

Its components satisfy $G_{0i} = -B_i$, $G^{ij} = -\varepsilon^{ijk}E_k/c$ for $i, j, k = 1, 2, 3$.

- **Levi-Civita symbol** in 3D satisfies

$$\varepsilon_{ijk} \;=\; \varepsilon^{ijk} \;=\; \begin{cases} +1 & \text{if } \; ijk \text{ is an even permutation} \\ -1 & \text{if } \; ijk \text{ is an odd permutation.} \end{cases}$$

Here, even permutations consist of $123, 231, 312$, and odd permutations are $132, 213, 321$. It also satifies $\varepsilon_{ijk}\varepsilon^{imn} = \delta_j^m\delta_k^n - \delta_k^m\delta_j^n$, and

$$\boldsymbol{a}\cdot(\boldsymbol{b}\times\boldsymbol{c}) \;=\; \varepsilon^{ijk}a_ib_jc_k.$$

- **Levi-Civita symbol** in 4D satisfies

$$\varepsilon_{\mu\nu\rho\sigma} \;=\; -\varepsilon^{\mu\nu\rho\sigma} \;=\; \begin{cases} +1 & \text{if } \; \mu\nu\rho\sigma \text{ is an even permutation of } 0123 \\ -1 & \text{if } \; \mu\nu\rho\sigma \text{ is an odd permutation of } 0123 \\ 0 & \text{if } \; \text{anything else.} \end{cases}$$

The signs of the permutations of $0, 1, 2, 3$ are given by

$+$	0123,	0231,	0312,	1032,	1203,	1320,	2013,	2130,	2301,	3021,	3102,	3210,
$-$	0132,	0213,	0321,	1023,	1230,	1302,	2031,	2103,	2310,	3012,	3120,	3201.

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.